INTERIM FEDERAL SPECIFICATION

CONDUIT, RACEWAY, METAL;

CONDUIT, RACEWAY, NONMETALLIC;

AND FITTINGS (UNDERFLOOR)

This Interim Federal Specification was developed by General Services Administration, Federal Supply Service, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers underfloor duct systems for wires and cables. This specification does not cover underfloor distribution systems of the cellular type.

1.2 Classification.

1.2.1 Types. Underfloor duct systems, furnished under this specification shall be of the following types, as specified (see 6.1):

Type I - Afterset insert

Type II - Preset insert

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Specifications:

PPP-B-601 Boxes, Wood, Cleated-Plywood

PPP-B-621 Boxes, Wood, Nailed and Lock-Corner

PPP-B-636 Boxes, Fiberboard

Federal Standards:

Fed. Std. No. 123 Marking for Domestic Shipment

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116 Preservation, Methods of

Military Standards:

MIL-STD-129 Marking for Shipment and Storage

MIL-STD-130 Identification Marking of U. S. Military Property

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Underwriters' Laboratories, Inc. Standard:

UL-884 Underfloor Raceways and Fittings

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, N. Y. 11749; 207 East Ohio Street, Chicago, Illinois 60611; or 1655 Scott Boulevard, Santa Clara, California 95050).

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Except as otherwise provided herein, the underfloor ducts and fittings covered by this specification shall conform to the applicable requirements of Underwriters' Laboratories, Inc., Standard UL-884. All boxes, couplings, elbows, bends, duct supports, etc., shall be suitable for use with the duct furnished to form the complete system.

3.2 Fire and casualty hazards.

3.2.1 Each contractor shall submit to the contracting agency proof that the underfloor ducts and fittings he proposes to supply under this specification conform to the requirements of UL-884. The label, or listing with re-examination, of the Underwriters' Laboratories, Inc., may be accepted as evidence that the underfloor ducts and fittings conform to this requirement.

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3.2.2 In lieu of the label, or listing with re-examination, the contractor may submit independent proof, satisfactory to the contracting agency, that the underfloor ducts and fittings conform to the applicable requirements of the published standards, including methods of tests, of the Underwriters' Laboratories, Inc., UL-884 Standard.

3.2.3 Compliance with the above preliminary requirements in regard to fire and casualty hazards does not absolve the contractor from complete compliance with the requirements of this specification in order to secure acceptance of his material.

3.3 Material. Unless otherwise specified (see 6.1), either metallic or nonmetallic underfloor duct material may be furnished under this specification.

3.4 Dimensions. The duct shall have an internal cross-sectional area of not less than 3 square inches (19.355 square centimeters). The minimum external height shall be 1-1/4 inches (3.175 centimeters). The length shall be in accordance with the manufacturers' standard practice.

3.5 Instructions. Clear and complete instructions shall be provided for the installation of the underfloor duct systems, and shall include illustrations of all essential features and the relation of component parts. Special attention shall be given to items which require particular care on the part of the contractors or workmen.

3.6 Special tools or equipment. If special tools or equipment are required for any of the operations incident to the proper installation of duct or fittings, such as locating afterset inserts, cutting through floor or into duct, clamping or fastening fittings in place, etc., such equipment shall be provided and the instructions shall explain clearly how to use them correctly. Quantities and disposition of special tools or equipment, and instructions, shall be furnished as specified (see 6.1).

3.7 Workmanship. The conduit and fittings shall be manufactured and finished in a thoroughly workmanship manner in accordance with all applicable requirements of this specification. They shall be free from all defects that could abrade or otherwise damage wire or cables.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling, inspection, and tests. Unless otherwise specified in the invitation for bids (see 6.1), sampling, inspection, and tests shall be in accordance with the applicable requirements of UL-884.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

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5.1.1 Level A. The underfloor ducts and fittings shall be packaged by method III of Mil-P-116.

5.1.2 Level C. The underfloor ducts and fittings shall be packaged in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be levels A, B, or C as specified (see 6.1).

5.2.1 Level A. The underfloor ducts and fittings shall be packed in overseas type cleated-plywood boxes conforming to PPP-B-601, or overseas class nailed wood boxes conforming to PPP-B-621. The gross weight of boxes shall not exceed approximately 200 pounds.

5.2.2 Level B. The underfloor ducts and fittings shall be packed in domestic type cleated-plywood boxes conforming to PPP-B-601, domestic class nailed wood boxes conforming to PPP-B-636. Contents of boxes shall be cushioned and braced to prevent movement within the boxes. The gross weight of wood boxes shall not exceed approximately 200 pounds; of fiberboard boxes, approximately 70 pounds.

5.2.3 Level C. The underfloor ducts and fittings shall be packed or prepared for shipment in a manner which will insure arrival at destination in satisfactory condition and which will be acceptable to the carrier at lowest rates.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required in the contract or purchase order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required in the contract or order, shipping containers shall be marked in accordance with Mil-Std-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, symbol, and date of this specification.

(b) Type of underfloor duct desired (1.2).

(c) Material, if specific material is required (3.3).

(d) Special tools, equipment and instructions required (3.6).

(e) Inspection responsibility, if other than specified (4.1).

(f) Sampling, inspection, and tests, if other than specified (4.2).

(g) Packaging and Packing level required (5.1 and 5.2).

(h) Special marking, if required (5.3).

6.2 Supersession data. It is the intent of this specification to supersede W-R-36c, Raceways and Fittings, Metallic, Underfloor, dated November 7, 1963.

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6.3 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

METAL

Rail:

Conduits or Conduit Connections, flexible, steel, for other than underground works
Carload minimum weight 36,000 pounds.

Motor:

Pipe, Conduit or Duct or Race-Ways.
Truckload minimum weight 40,000 pounds, subject to Rule 115, National Motor Freight Classification.

NON METALLIC

Rail:

Conduits or Conduit Connections, flexible, other than steel, other than underground works.
Carload minimum weight 30,000 pounds.

Motor:

Conduits, electrical, other than steel.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

FITTINGS

Rail:

Pipe fittings, not otherwise indexed by name, iron or steel.
Carload minimum weight 40,000 pounds.

Motor:

Pipe fitting, not otherwise indexed, iron or steel.
Truckload minimum weight 40,000 pounds, subject to Rule 115, National Motor Freight Classification.

INTERESTS:

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